

IAN THOMAS WHITE

Curriculum Vitae & Extended Dossier

Researcher | Software Engineer | Computational Linguist

ianthomaswhite[at]gmail.com | ianthomaswhite.com | github.com/ianthomaswhite

RESEARCH FOCUS & SUMMARY

Multidisciplinary researcher and software developer working at the intersection of computational linguistics, functional programming (Haskell), and artificial intelligence. Specializing in comparative Indo-European phonology, historical sound laws, and type-safe systems with strong formal guarantees.

EDUCATION

Bachelor of Arts / Science in Linguistics & Computer Science **2018 – 2022**

Academic Honors | Departmental Distinction

- Focus in historical phonology, comparative Indo-European grammar, and programming language theory.
- Independent research thesis on comparative sound reconstruction and phonetic data representations.
- Coursework: Formal Language Theory, Compilers, Type Systems, Comparative Philology, Phonetics & Phonology.

PROFESSIONAL & RESEARCH EXPERIENCE

Lead Developer & Computational Linguist **2023 – Present**

Independent Linguistics & AI Research

Remote / Independent

- Architected an IPA-based Indo-European comparative dictionary engine modeling phonological shift across Indo-European branches.
- Implemented static site generators and type-safe web services using Haskell (Hakyll, Yesod, Servant).
- Created automated LaTeX-to-web publishing pipelines with continuous delivery on GitHub Actions.
- Built tooling for automated grammatical verification and IPA alignment.

Research Engineer / Assistant

Academic & Technical Labs

2021 – 2023

United States

- Authored technical monographs, grammar specifications, and research reports in LaTeX.
- Engineered data ingestion pipelines for large lexical and textual corpora in Haskell and Python.
- Designed structured schemas for phonetic inventories and historical etymological networks.
- Collaborated with multidisciplinary teams on computational NLP benchmarks.

MAJOR PROJECTS

IPA Comparative Indo-European Lexicon & Engine **2024 – Present**

- Systematic reconstruction tool tracing Proto-Indo-European etyma across Germanic, Celtic, Romance, Slavic, Indo-Iranian, and Greek, strictly aligned to IPA representations.
- Implemented algorithmic validation for known historical sound shifts (Grimm's Law, Verner's Law, Grassman's Law).

Automated AI Resume & Dossier Generator

2025

- Programmatic pipeline transforming structured metadata into pixel-perfect PDF publications and custom LaTeX dossiers.

Haskell Brutalist Web Platform

2026

- Custom Hakyll static generator featuring zero-runtime dependencies, compile-time LaTeX PDF synthesis, and web-based publishing workflows.

PUBLICATIONS & WORKING PAPERS

- **"Proto-Indo-European Laryngeals and Vowel Coloring in Anatolian and Greek"** — Working Paper, 2025.
- **"On Formal Systems and Expressive Rigor in Functional Software"** — Theoretical Essay, 2024.
- **"Comparative Sound Laws of Balto-Slavic Accentuation"** — Research Monograph, 2023.

TECHNICAL COMPETENCIES

- **Languages:** Haskell, Bash, Python, SQL, HTML5/CSS3.
- **Typesetting & Publishing:** LaTeX (pdf_lat_ex, lua_lat_ex, TikZ), Pandoc, Markdown, Git.
- **Linguistic Methodology:** Historical Phonetics, Proto-Indo-European, IPA Transcription, Morphology.
- **Systems & Infrastructure:** Linux/Unix, GitHub Actions, Docker, Static Site Generation.